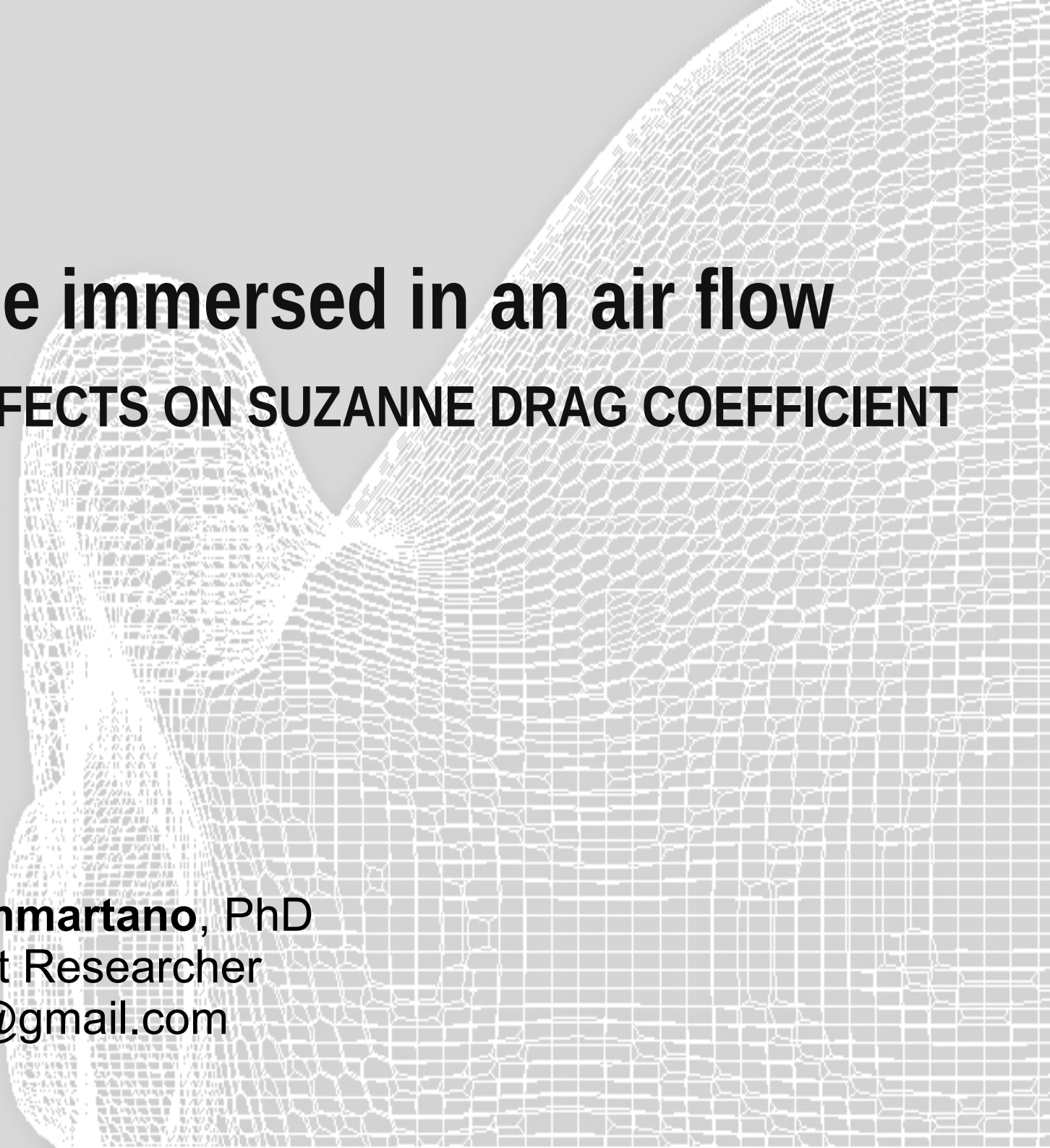




Suzanne immersed in an air flow

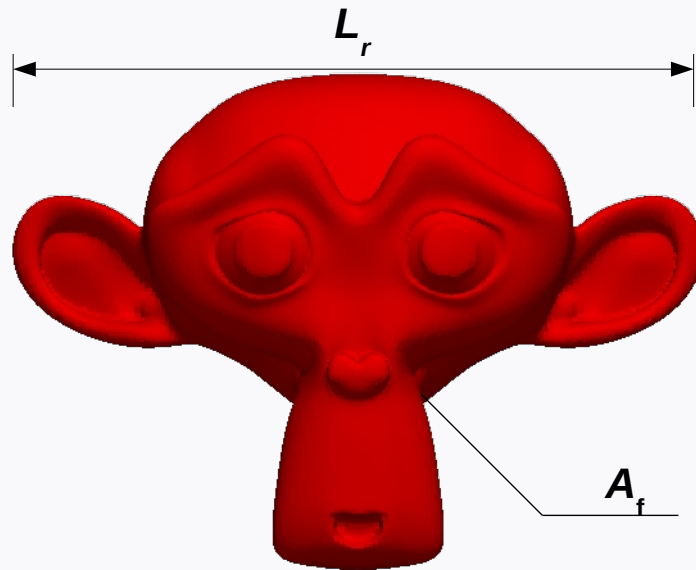
VELOCITY EFFECTS ON SUZANNE DRAG COEFFICIENT

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WorkFlow and CFD choices



- Removed the holes on Suzanne head by means of Blender;
- Estimation of the projected surface area A_f of Suzanne normal to the direction of flow (Paraview);
- Selection of the reference length L_r for Reynolds number estimation;



• $A_f = 2.4667 \text{ m}^2$;
• $L_r = 2.7 \text{ m}$;

- Selection of fluid type in its normal conditions (NTP):

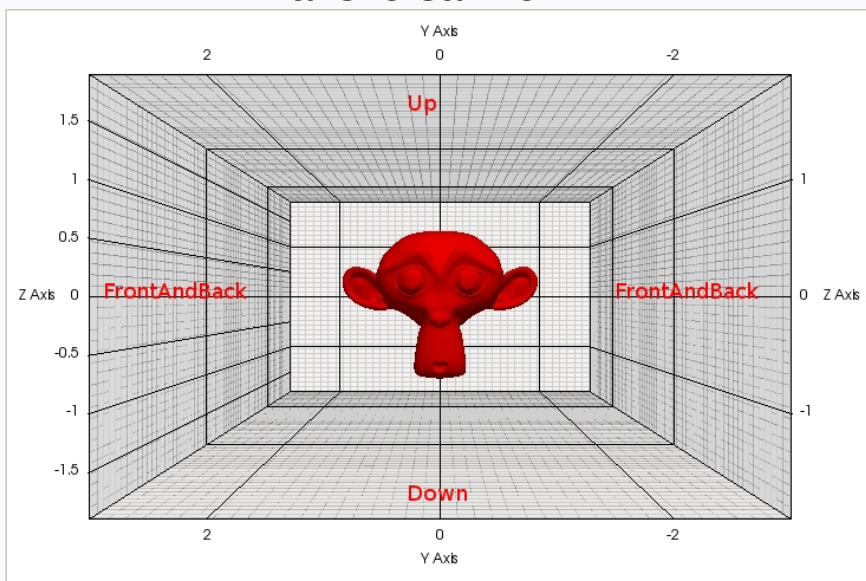
Air in normal conditions

Variable	Value	Unit
Temperature - T	20.00	°C
Absolute pressure - p_{atm}	1.00	atm
Density - ρ	1.204	kg/m ³
Specific Volume - γ	11.81	N/m ³
Dynamic Viscosity - μ	$1.72e^{-5}$	Pa s
Kinematic Viscosity - ν	$1.43e^{-5}$	m ² /s

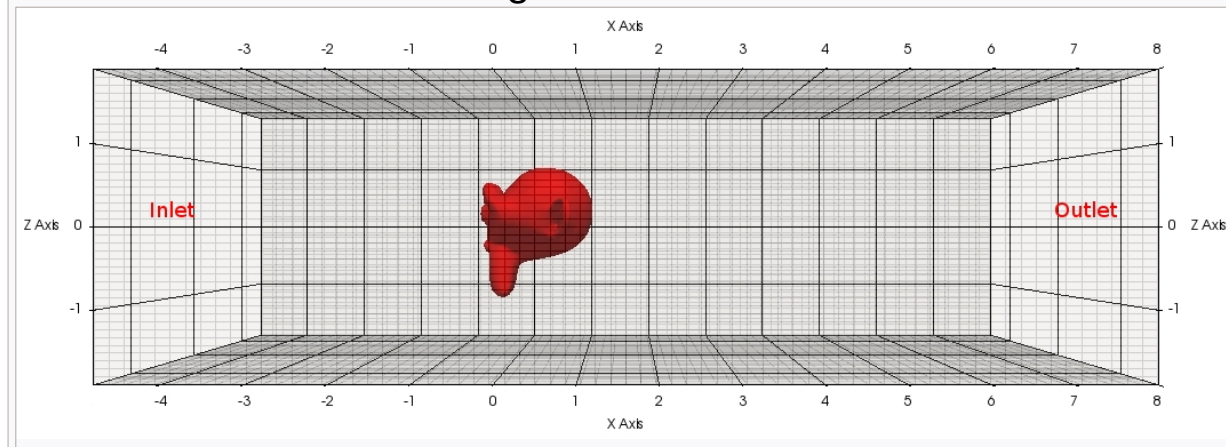


- Domain discretization by using blockMesh and snappyHexMesh:
 - ✓ **blockMesh** generates a base mesh with an **aspect ratio** ≈ 1 ;
 - ✓ **snappyHexMesh** discretizes the fluid domain around Suzanne's head;
 - ✓ Each run has been performed with a mesh such that $y^+ < 5$ over Suzanne's head;
 - ✓ Each mesh has a number of cells strictly dependent on the stream velocity value U_∞ .
- The computational domain is a parallelepiped 12.8 m long, 6 m wide and 3.8 m tall:
 - Longitudinal direction : X Axis [-4.8, 8.0]
 - Vertical direction : Z Axis [-1.9, 1.9]
 - Transversal direction : Y Axis [-3.0, 3.0]

Transversal view

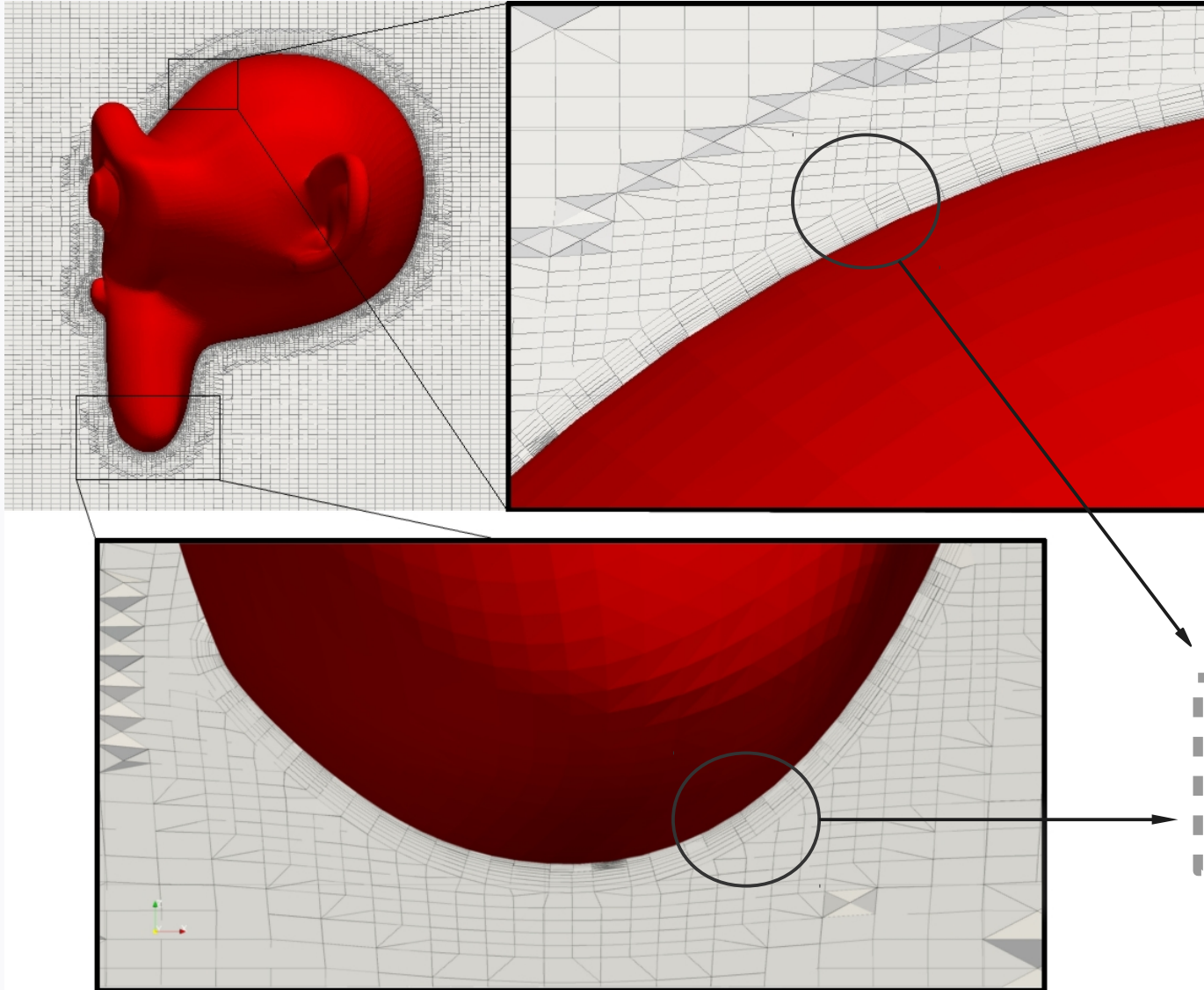


Longitudinal view





- Example of a mesh generated with snappyHexMesh for the case study with $U_{\infty} = 0.1 \text{ m/s}$



Mesh Statistics:

- Cells: 2,770,230
- Max aspect ratio: 37.9
- Max skewness: 3.99
- y^+ min : $5.0e-7$
- y^+ max: 3.191
- y^+ avg : 0.059

Multiple surface layers:

- Number of layers = 10
- Expansion Ratio = 1.2
- Final Layer Thickness = 0.4



- The drag coefficient C_d has been estimated for a wide range of the stream velocity $U_\infty = 1e^{-4} \div 5e^{+1}$ m/s
- CFD simulations settings:
 - ✓ Category: **RANS;**
 - ✓ Closure model: **SST k- ω turbulence Model;**
 - ✓ steady-state solver: **simpleFoam;**
 - ✓ unsteady-state solver: **pimpleFoam;**
- For the considered stream velocity range the **drag coefficient** C_d was esistimated as follow:

$$C_d = \frac{(2 \cdot F_d)}{(\rho_{air} \cdot A_f \cdot U_\infty^2)}$$

being F_d the drag force.

- Setting up the boundary conditions **BCs** according to the following table:

Patch	Vars	U	p	k	nut	ω
Inlet		freestream	freestreamPressure	fixedValue	calculated	fixedValue
Outlet		freestream	freestreamPressure	inletOutlet	calculated	inletOutlet
Face		noSlip	zeroGradient	kqRWallFunction	nutkWallFunction	omegaWallFunction
Down Up		freestream	freestreamPressure	inletOutlet	calculated	inletOutlet
frontAndBack		freestream	freestreamPressure	inletOutlet	calculated	inletOutlet

Results

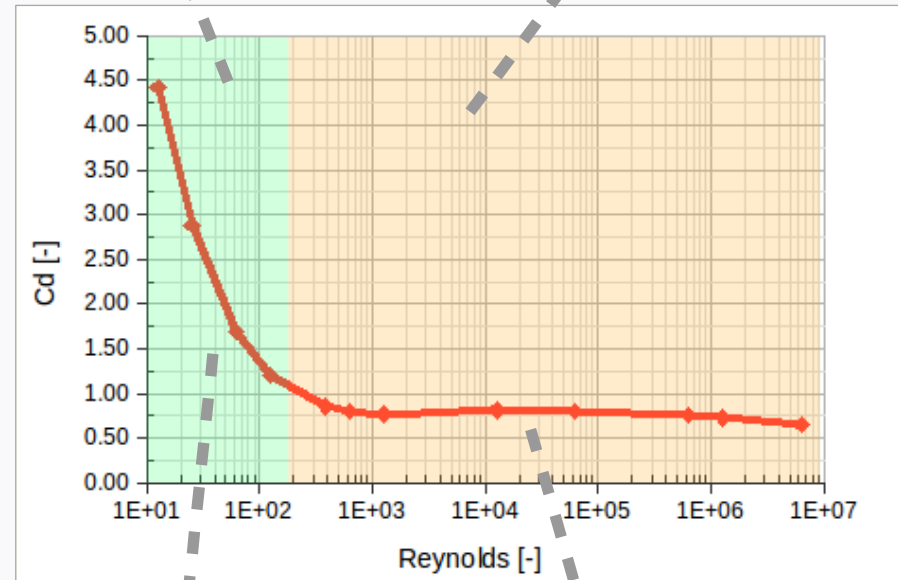


- Table and plot of the drag coefficient C_d values relative to different Reynolds number values.

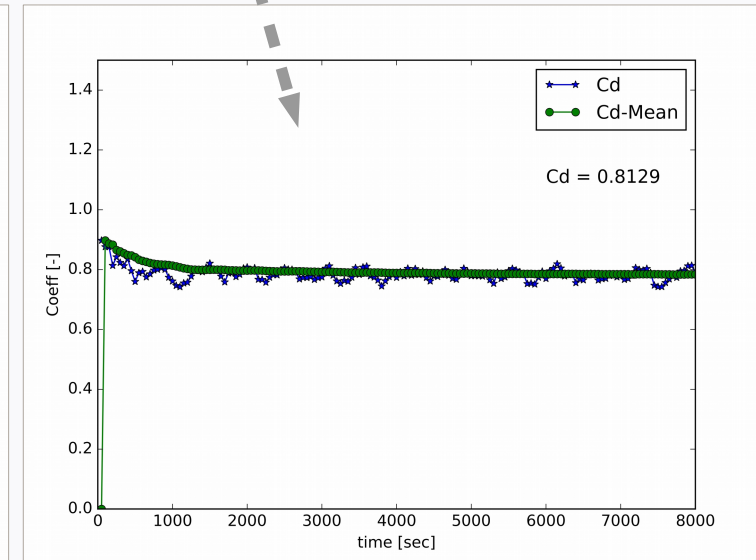
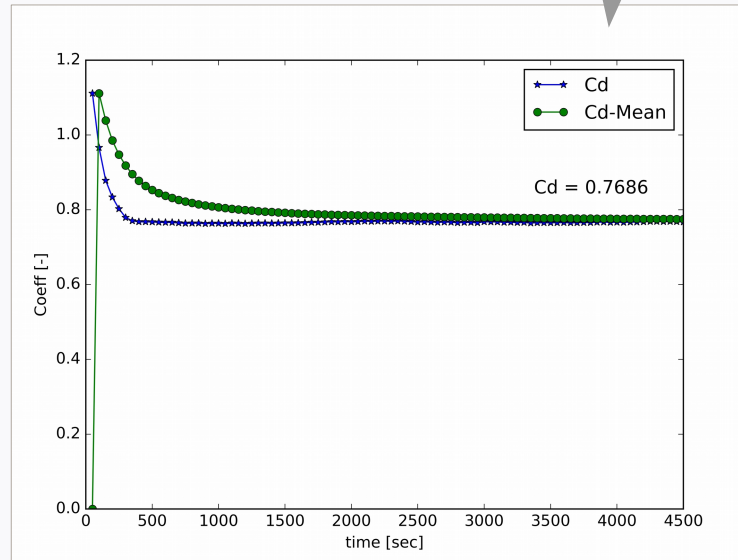
U_∞ [m/s]	Reynolds [-]	C_d [-]
0.0001	1.89E+01	4.4211
0.0002	3.78E+01	2.8768
0.0005	9.45E+01	1.6932
0.001	1.89E+02	1.2085
0.003	5.67E+02	0.8547
0.005	9.45E+02	0.7937
0.01	1.89E+03	0.7686
0.1	1.89E+04	0.8129
0.5	9.45E+04	0.7949
5	9.45E+05	0.7604
10	1.89E+06	0.7288
50	9.45E+06	0.6558

Steady-state regime

Unsteady-state regime



- Time series of the drag coefficient C_d of two simulations in different flow regime.



Results

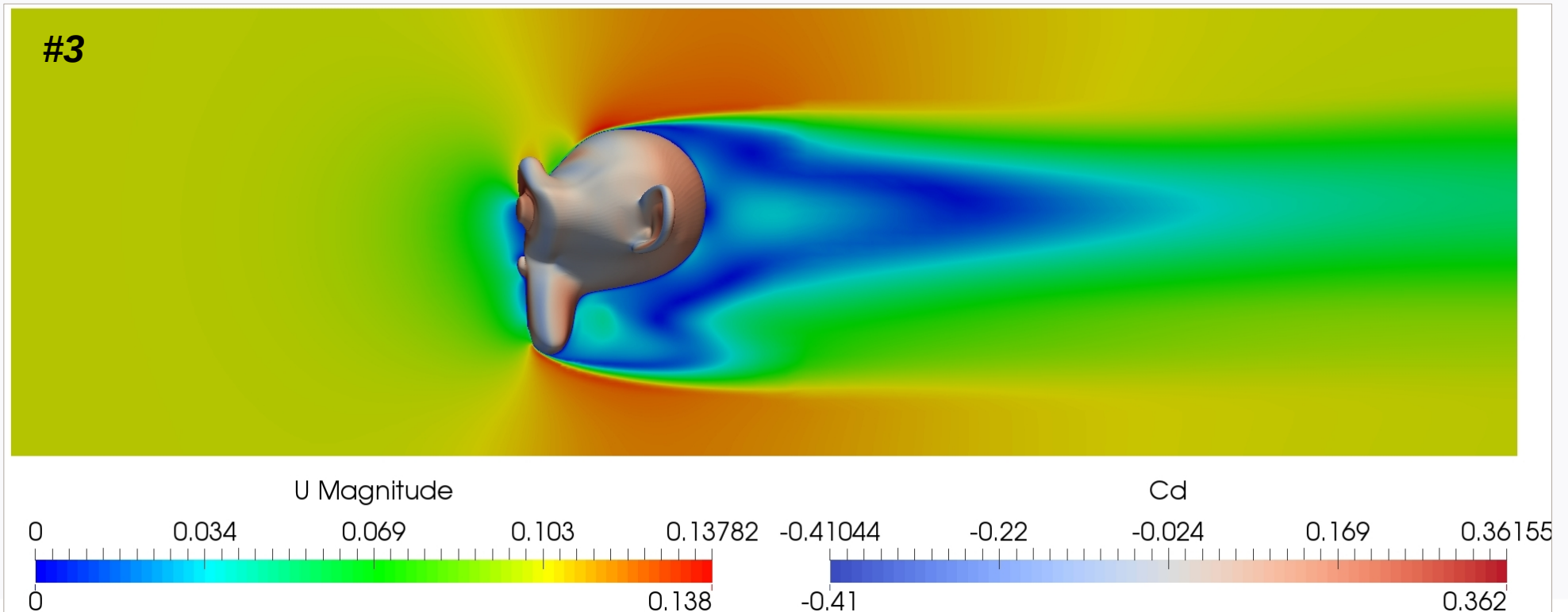
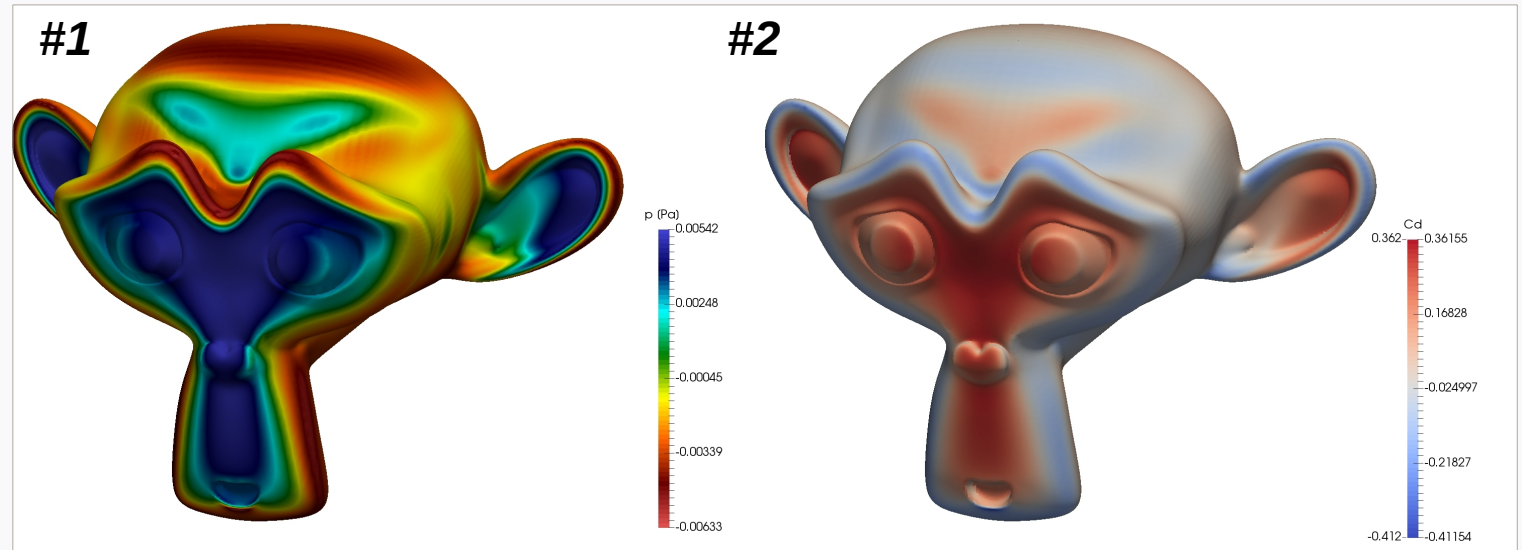


- Contour plots of pressure, drag coefficient and velocity magnitude for the case study with $U_{\infty} = 0.1 \text{ m/s}$

#1 Pressure

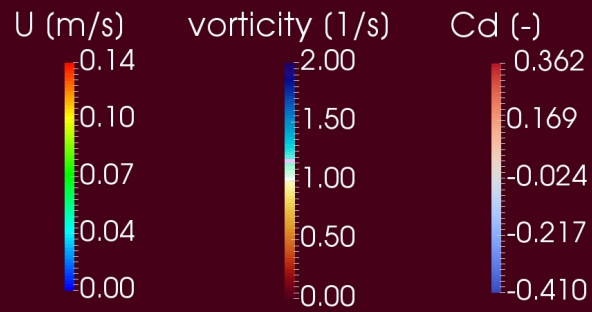
#2 Cd

#3 U magnitude

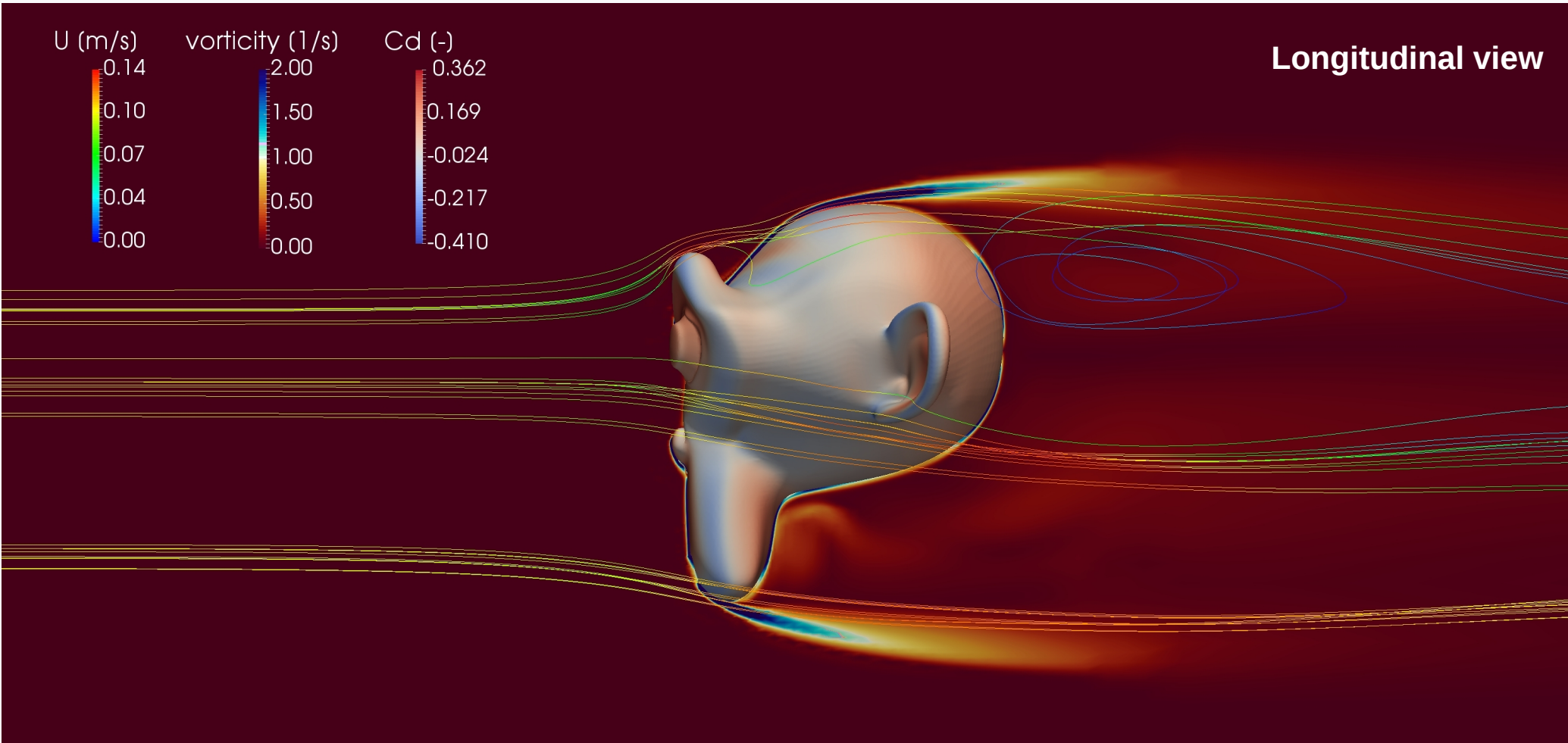




- Contour plot of vorticity and stream lines for the case study with $U_{\infty} = 0.1 \text{ m/s}$ - Longitudinal view



Longitudinal view





- Contour plot of vorticity and stream lines for the case study with $U_{\infty} = 0.1 \text{ m/s}$ – Frontal view

